

HIGHER SCHOOL CERTIFICATE EXAMINATION

2022

Mathematics Advanced

Examiner: Sami El Hosri

General Instructions

- Reading Time – 10 minutes
- Working Time – 3 hours
- Write using black pen
- Calculators approved by NESAC may be used
- A separate reference sheet is provided at the back of this paper.
- All necessary working should be shown in Questions 11 – 36.

Total marks – 100

Section I: Multiple Choice

Questions 1 – 10 10 marks

- Attempt all questions
- Allow about 15 minutes for this section.

Section II: Extended Response

Questions 11 – 35 90 marks

- Attempt all questions
- Allow about 2 hours 45 minutes for this section.

Section I

Questions 1 – 10 (10 marks)

Read each question and choose an answer A, B, C or D.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10.

- 1 The set of ordered pairs is a relation

$$\{(-2, 3), (-4, 5), (3, 2), (-2, 1), (7, 2)\}.$$

Which of the following types is this relation?

- A) one to one
 - B) one to many
 - C) many to one
 - D) many to many
- 2 The function $g(x) = 4 \cos 2x$ is to be dilated horizontally by a factor of $\frac{1}{4}$ and then translated horizontally $\frac{\pi}{6}$ units to the right.
- What is the equation of the transformed function $h(x)$?

A) $h(x) = 4 \cos 8 \left(x - \frac{\pi}{6} \right)$

B) $h(x) = 4 \cos \frac{1}{2} \left(x - \frac{\pi}{6} \right)$

C) $h(x) = 4 \cos \left(\frac{1}{2} x - \frac{\pi}{6} \right)$

D) $h(x) = 4 \cos \left(8x - \frac{\pi}{6} \right)$

- 3 Which of the following is the range of the function $g(x) = 1 + 2\sqrt{4 - x^2}$?

A) $[1, 3]$

B) $[0, 2]$

C) $[1, 5]$

D) $[0, 5]$

- 4 A jar contains 30 seashells. 19 of these shells are white and 11 are black.

Kate is to select two shells from the jar at random.

What is the probability that Kate selects two white shells?

A) $\frac{19}{30} \times \frac{18}{29}$

B) $\frac{19}{30} + \frac{18}{30}$

C) $\frac{19}{30} + \frac{18}{29}$

D) $\frac{19}{30} \times \frac{18}{30}$

- 5 David is playing a game of chance. He is to select one card from a bag at random. The cards in the bag are as shown.

5	4	1	3	2	6	4	6
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The table shows the results of selecting various cards.

Card	Result
6	Win \$40
3, 4, 5	Win \$20
1, 2	Lose \$32

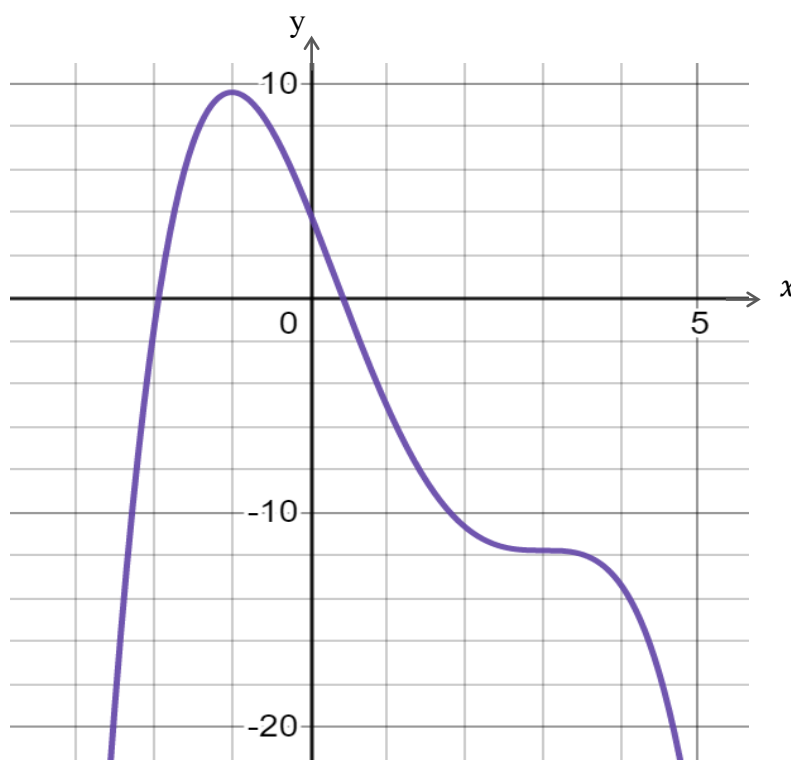
Given that David must pay \$10 to play the game, what should David's financial expectation be when he plays this game once?

- A) Gain \$12
- B) Gain \$2
- C) Lose \$2
- D) Gain \$18

- 6 In a class test, the students' marks were between 70 and 90. The class teacher calculates the mean and the standard deviation for the students' marks in this class test.
- Luke is a student in this class but was absent. He did the test later and his mark was 50.
- If Luke's mark is to be included, what effect would it have had on the mean and the standard deviation of the students' marks in this class test?

- A) Both the mean and the standard deviation will decrease.
- B) The mean will decrease and the standard deviation will stay the same.
- C) The mean will decrease and the standard deviation will increase.
- D) Both the mean and the standard deviation will stay the same.

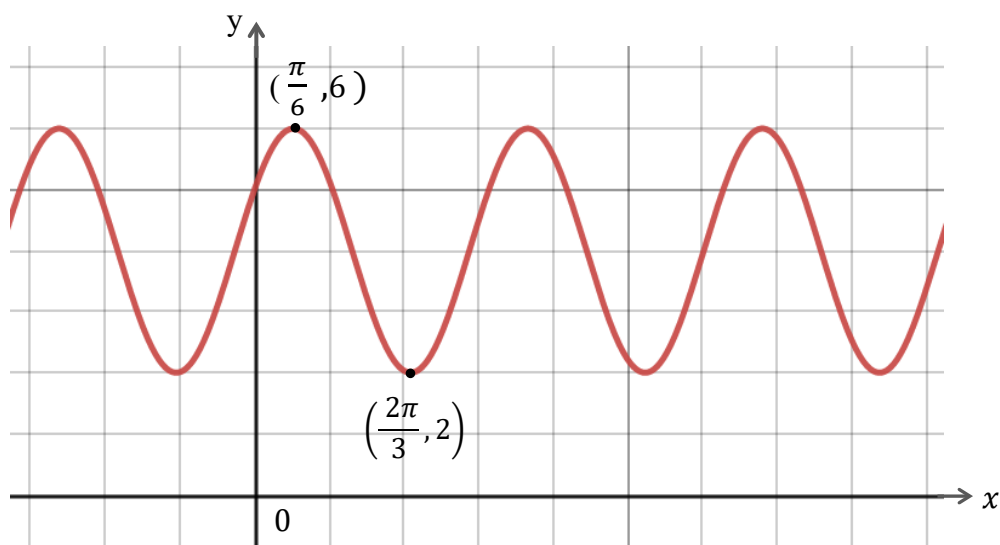
- 7 The diagram shows the graph of $y = h(x)$.



Which of the following could be the equation of $h'(x)$?

- A) $h'(x) = -(x + 1)(x - 3)^2$
- B) $h'(x) = (x + 1)(x - 2)(x - 4)$
- C) $h'(x) = (x + 1)(x - 3)^2$
- D) $h'(x) = (x + 1)(x - 2)(4 - x)$

- 8 Which of the equations below is represented by the graph?



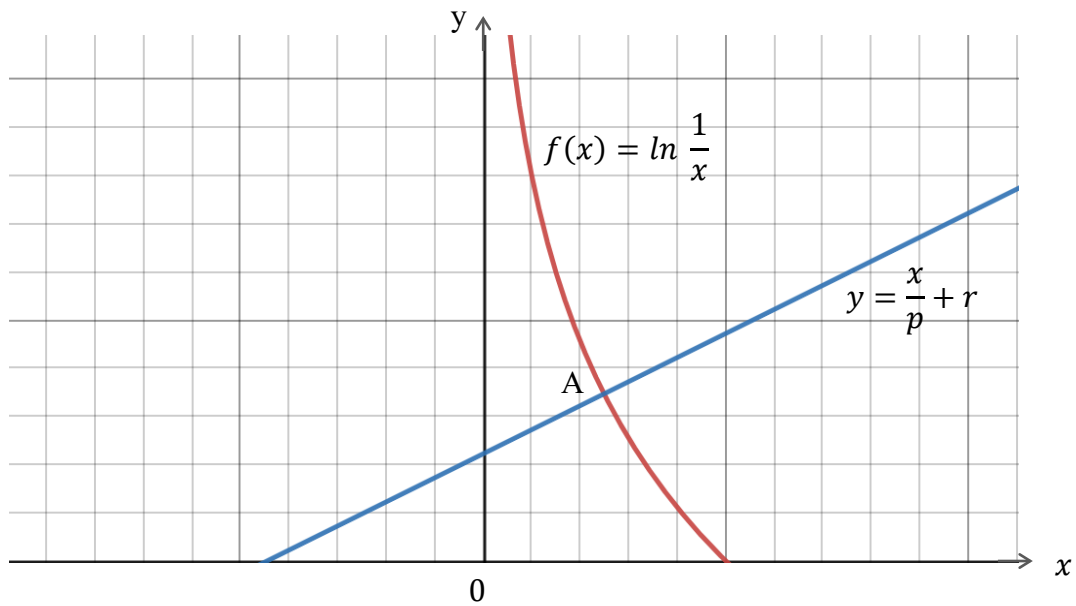
- A) $y = 4 + 2 \cos\left(2x - \frac{\pi}{4}\right)$
 B) $y = 4 + 2 \cos\left(2x - \frac{\pi}{3}\right)$
 C) $y = 4 + 2 \cos\left(2x + \frac{\pi}{6}\right)$
 D) $y = 4 + 2 \cos\left(2x - \frac{\pi}{6}\right)$
- 9 The functions $F(x)$, $G(x)$ and $H(x)$ are defined for all real x and the function

$$H(x) = F\left(\frac{1}{G(x)}\right).$$

Given that $G(3) = \frac{1}{2}$, $G'(3) = \frac{1}{10}$ and $F'(2) = 5$, what is the gradient of the tangent to the curve $H(x)$ at $x = 3$?

- A) 20
 B) -20
 C) 2
 D) -2

- 10 Consider the part of the function $f(x) = \ln \frac{1}{x}$ in the first quadrant.
The normal at the point A with x coordinate q has equation $y = \frac{x}{p} + r$
as shown in the diagram below.



Which of the following inequalities is correct?

- A) $\ln q < p < q$
- B) $\ln q < q < p$
- C) $q < p < \ln q$
- D) $q < \ln q < p$

ADVANCED MATHEMATICS

Section II Answer Booklet 1

Section II

90 marks

Attempt Questions 11 – 36

Allow about 2 hours and 45 minutes for this section.

Booklet 1 - Attempt Questions 11 – 30 (64 marks)

Booklet 2 - Attempt Questions 31 – 35 (26 marks)

INSTRUCTIONS

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space for Booklet 1 of Section II is provided on pages 20 and 21.
Extra writing space for Booklet 2 of Section II is provided on pages 29 and 30.

If you use this space, clearly indicate which question you are answering.

Question 11 (2 marks)

Find the coordinates of the centre and the radius of the circle with equation

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$$x^2 - 10x + y^2 + 4y + 1 = 0.$$

Question 12 (2 marks)

Find the exact gradient of the tangent to the curve $y = \sin x$ at the point where $x = \frac{\pi}{6}$.

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Question 13 (4 marks)

Consider the functions $f(x) = \sqrt{2x+1}$ and $g(x) = 2x - 3$.

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(a) Find $f(g(x))$.

(b) Find the value of x given that $f(g(x)) = 3$.

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Question 14 (2 marks)

Evaluate $\int_1^2 \frac{2}{x} dx$

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Question 15 (2 marks)

Evaluate $\int_0^{\frac{\pi}{2}} \cos 2x \, dx$

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Question 16 (2 marks)

Solve $\log_2 x = 6 - \log_2 x$

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Question 17 (2 marks)

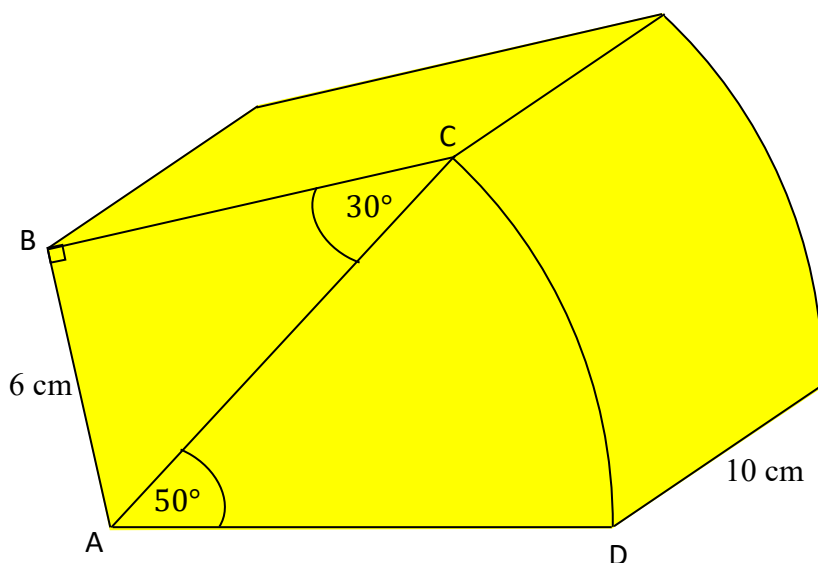
A and B are two mutually exclusive events.

If $P(B) = 2P(A)$ and $P(A \cup B) = \frac{4}{9}$, what is the value of $P(A)$?

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Question 18 (6 marks)

The cross section of the prism shown in the diagram consists of the right-angled triangle ABC and the sector ACD.



(a) Find the length of the radius AC of the sector.

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(b) Find the exact area of the sector.

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(c) Hence, find the volume of the prism correct to two decimal places.

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Question 19 (4 marks)

Last term, a group of students spent different numbers of hours studying for their Mathematics yearly exam.

For each student the time, in hours, spent on studying in the last term and the marks scored in the Mathematics yearly exam were recorded.

The table shows the mean and standard deviation for the study hours of all the students in the group.

	Mean	Standard deviation
Time in hours spent on studying by the students in the group ($\bar{x}$)	76.4	16.66

The equation of the least squares line of best fit, relating the time in hours spent by students studying (x) and their score in the yearly exam (y), has been calculated to be

$$y = 46.35 + 0.539x \text{ and the correlation coefficient is } r = 0.971.$$

- (a) Based on this equation, if a student scored 90 in the yearly exam, what would be the time spent by this student on studying for the test in the last term?

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- (b) The equation of the least squares line is $y = \text{gradient} \times x + y \text{ intercept}$, such that $\text{gradient} = r \times \frac{\text{standard deviation of } y}{\text{standard deviation of } x}$ and $y \text{ intercept} = \bar{y} - \text{gradient} \times \bar{x}$

Use the information above to calculate the standard deviation and the mean of the students in their Mathematics yearly exam.

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Give your answers to two decimal places.

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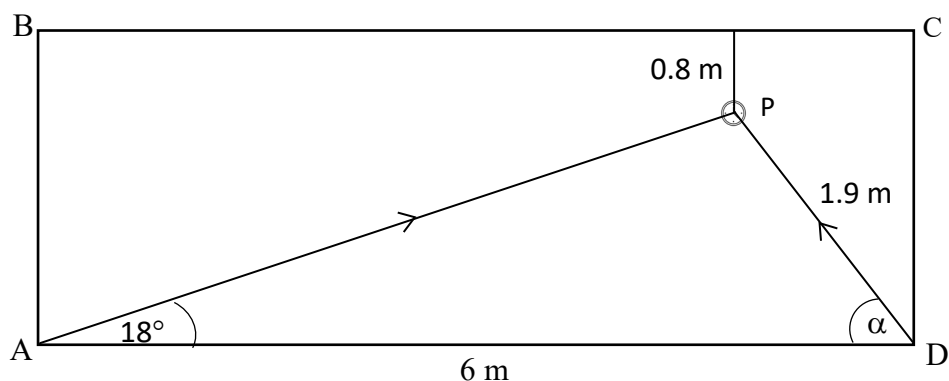
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Question 20 (4 marks)

In the diagram ABCD, is the back wall of a room 6 m long. A clock is hanging at a point, P, on this wall. Its distances from the edge BC of the ceiling and from the corner D on the ground are 0.8 m and 1.9 m respectively.

The angles of elevations of this clock from the corners A and D on the ground are 18° and α respectively and angle APD is obtuse.



- (a) Find α the angle of elevation of the clock P from D correct to the nearest minute.

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- (b) Find AB the height of the wall in this room correct to two decimal places.

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QUESTION 21 (2 marks)

For what values x is the curve $f(x) = -x^3 + x^4$ concave up?

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QUESTION 22 (3 marks)

Without using calculus, sketch the graph of the curve $y = \frac{x-2}{x+2}$ showing the asymptotes and the points of intersection with the x and the y axes.

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QUESTION 23 (4 marks)

The price per gram of 18 carat gold sold on an island in jewellery shops at different towns is found to be normally distributed with a mean of \$40 and standard deviation of \$2.

- (a) Find the probability that a jewellery shop sells one gram of 18 carat gold between \$34 and \$38.

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- (b) If the selling price of only one jewellery shop out of every 40 jewellery shops is in the top range of the prices, what is the lowest selling price in the top range of prices in these shops?

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QUESTION 24 (2 marks)

Find the coordinates of the points of intersection between the line $y = 2$ and the curve $y = 4 \sin 2x$ in the domain $0 \leq x \leq \frac{\pi}{2}$.

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QUESTION 25 (3 marks)

The table below shows the future value of annuity with a contribution of \$1 at the end of each period.

Interest rate per period							
Period	0.75%	1.00%	1.50%	2.00%	2.50%	3.00%	4.00%
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0075	2.0100	2.0150	2.0200	2.0250	2.0300	2.0400
3	3.0226	3.0301	3.0452	3.0604	3.0756	3.0909	3.1216
4	4.0452	4.0604	4.0909	4.1216	4.1525	4.1836	4.2465
5	5.0756	5.1010	5.1523	5.2040	5.2563	5.3091	5.4163
6	6.1136	6.1520	6.2296	6.3081	6.3877	6.4684	6.6330
7	7.1595	7.2135	7.3230	7.4343	7.5474	7.6625	7.8983
8	8.2132	8.2857	8.4328	8.5830	8.7361	8.8923	9.2142

Kayla opened an account and deposited \$5 000 at the end of a year and made no other deposits for 4 years. The money in her account was earning interest at 2.5% per annum compounded annually.

At the end of the 5th year, she deposited \$2 000 in the same account and continued to deposit \$2 000 at end of each year until the end of the 8th year.

How much money will Kayla have in her account at the end of 8 years?

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This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 26 (4 marks)

The height of a tree in a park is modelled by $h = 750 - 650 a^{-0.5t}$, where h is the height of the tree in centimetres (cm) and t is its age in years since the tree was planted in the park.

The tree is 490 cm high after 2 years.

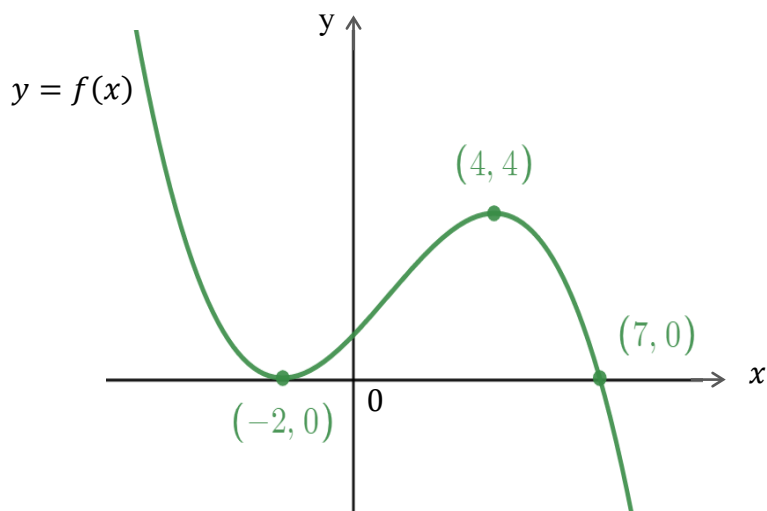
Find the value of t , correct to one decimal place, for which the instantaneous rate of increase is 12 cm per year.

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[illegible]

Question 27 (3 marks)

The diagram shows the graph of the function $y = f(x)$.



Sketch the graph of the function $y = -2f\left(\frac{x}{2}\right)$ showing the coordinates of the turning points and the x intercepts.

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[illegible]

The function $y = \frac{1}{6}(5x^3 - x^4)$ intersects the parabola $y = \frac{4}{9}(x - 5)^2$ at the points $(2, 4)$ and $(5, 0)$.

A Cartesian coordinate system showing two functions, $f(x) = x^2 - 5x + 6$ and $g(x) = x^2 - 2x$. The x-axis and y-axis are shown. The function $f(x)$ is a parabola opening upwards with x-intercepts at $(0, 0)$ and $(5, 0)$, and a vertex at $(2.5, -2.25)$. The function $g(x)$ is a parabola opening upwards with x-intercepts at $(0, 0)$ and $(2, 0)$, and a vertex at $(1, -1)$. The region between the two curves from $x=0$ to $x=5$ is shaded green. The intersection point $(2, 4)$ is marked with a black dot and labeled. The x-intercept of $g(x)$ at $(5, 0)$ is also marked with a black dot.

4

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 29 (6 marks)

A particle starts to move from the origin along the x axis. Its velocity at a time t seconds is given by

$$v = 4 - \frac{2}{3t+1}$$

- (a) Explain why the particle is never at rest and find its limiting velocity.

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- (b) Find the acceleration of the particle after 3 seconds.

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- (c) Find the exact distance travelled by the particle between t = 0 and 5 seconds.

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Section II extra writing space

This space can be used if needed. Ensure that you label any answers written here clearly giving their question number and part letter.

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This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

MATHEMATICS ADVANCED

Section II Answer Booklet 2

Booklet 2 - Attempt Questions 31 – 35 (26 marks)

INSTRUCTIONS

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space for Booklet 1 of Section II is provided on pages 20 and 21.
Extra writing space for Booklet 2 of Section II is provided on pages 29 and 30.

If you use this space, clearly indicate which question you are answering.

Question 31 (5 marks)

Kevin decided to set up a trust fund for his grandson David.

He deposited \$P on the day that David was born and continued to deposit the same amount on each of David's birthdays until, and including, his 21st birthday.

The money is invested at 4.2 % per annum compounded monthly and can be withdrawn on David's 21st birthday.

- a) What will be the value of his first deposit on David's 21st birthday in terms of P?

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- b) What is the value of P if the total value of the fund is \$35 389 on David's 21st birthday?

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Question 32 (5 marks)

A company produces bottles of milk. The volumes of milk in the bottles are normally distributed with a mean μ mL and a standard deviation of 9 mL.

One bottle produced by this company is selected at random its volume is 286 mL and its

z score is $\frac{2}{3}$.

(a) Find the mean μ of these bottles.

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(b) Two of the bottles produced by this company are to be selected at random.

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Find the probability that at least one of them has a volume between 289 mL and 298 mL.

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Question 33 (6 marks)

A continuous random variable, X , has the following probability density function.

$$f(x) = \begin{cases} \frac{e}{8} \left(-4e^{-\frac{x}{2}} + px + 5 \right) & \text{for } 0 \leq x \leq 2 \text{ and } p \text{ is a constant.} \\ 0 & \text{for all other values of } x \end{cases}$$

(a) Given that the mode is $2 \ln 2$, find the value of p .

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(b) Show that the probability $P(0 \leq x \leq 1) = \sqrt{e} - \frac{7e}{16}$.

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(c) Given that a score is more than 1, what is the probability that the score is less than the median? Give your answer to 5 decimal places.

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Question 34 (4 marks)

A $(2e, 1)$ lies on the function $h(x)$. The tangent ℓ to $h(x)$ at A has equation $y = \frac{x}{2e}$.
B is the image of the point A on the function $g(x) = 3h(2x + 4)$.

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Find the equation of the tangent to $g(x)$ at B.

[illegible]

Question 35 (6 marks)

A bag contains only strawberry and honey lollies. There are at least 4 lollies of either of the two flavours and the total number of lollies in the bag is n .

Three lollies are to be selected at random from the bag without replacement.

Let X be the number of strawberry lollies selected.

The probability distribution table for the number X of strawberry lollies selected is as shown.

x	0	1	2	3
$P(X)$	$\frac{b}{9}$	b	$\log_9 a$	$\frac{5b}{9}$

(a) Show that $0 \leq b \leq \frac{3}{5}$

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(b) Given that $E(X) = \frac{9}{5}$, find the values of a and b .

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Question 35 continues on page 28

- (c) Given that the total number of lollies in the bag is $n = 10$, use guess and check to find, the number of strawberry lollies in the bag.

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END OF PAPER

[illegible]

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**Mathematics Advanced**
Mathematics Extension 1
Mathematics Extension 2**REFERENCE SHEET****Measurement****Length**

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

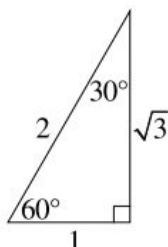
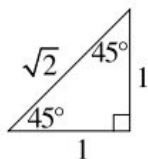
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

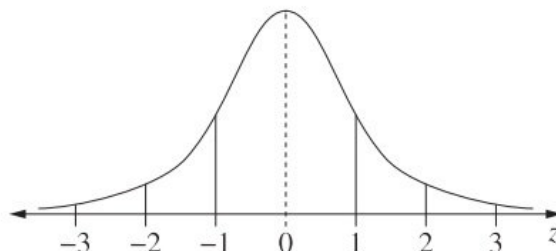
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= {}^n C_x p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x_1\underline{i} + y_1\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$z = a + ib = r(\cos \theta + i \sin \theta) \\ = re^{i\theta}$$

$$\left[r(\cos \theta + i \sin \theta) \right]^n = r^n(\cos n\theta + i \sin n\theta) \\ = r^n e^{in\theta}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$